# CLARA DIMARCO

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## RESEARCH INTERESTS

**Human-Computer Interaction and AI in Education**. My research addresses the measurement gap in adaptive learning environments. Standard automated metrics measure correctness but miss usability and accessibility barriers in LLM-generated logic proofs and tutoring feedback. I build frameworks that make those barriers measurable, through tutoring-system evaluation methods and co-design alongside learners with disabilities.

## PUBLICATIONS

### Journal Articles

[1] Sutapa Dey Tithi, Arun Kumar Ramesh, **Clara DiMarco**, Xiaoyi Tian, Nazia Alam, Kimia Fazeli, and Tiffany Barnes. 2025. The Promise and Limits of LLMs in Constructing Proofs and Hints for Logic Problems in Intelligent Tutoring Systems. Computers and Education: Artificial Intelligence 9 (2025), 100490. https://doi.org/10.1016/j.caeai.2025.100490

### Conference Proceedings

[2] Xiaoyi Tian, Yasitha Rajapaksha, Ally Limke, **Clara DiMarco**, Emily Bryans Dobar, Marnie Hill, Jamie Payton, and Tiffany Barnes. 2026. AI Scholars Program: Scaling AI Literacy Through K-12 Outreach. In Proceedings of the AAAI Conference on Artificial Intelligence (EAAI Track), Vol. 40. 40704-40712. https://doi.org/10.1609/aaai.v40i47.41518

## PRESENTATIONS

**DiMarco, C**., Tithi, S. D., Ramesh, A., and Barnes, T. Evaluating Prompting Techniques for Logic Proof Generation in GPT-4: Introducing Bidirectional Prompting. Poster.

- *NAE Grand Challenges Scholars Program International Annual Meeting*, NC State University, February 2025.
- *Grand Challenges Scholars Program Fall Symposium*, NC State University, Fall 2024.
- *NCSU Summer Research Symposium*, Summer 2024.

**DiMarco, C**., Hermida, V., Mantravadi, A., and Yakout, Z. Inclusive AI Learning: Promoting Accessibility in Computing Education for K-12 Students. Poster presentation, STARS Celebration Conference, 2024.

**Keynote speaker**, NAE Grand Challenges Scholars Program International Annual Meeting, NC State University, February 2025.

**Invited panelist**, Summer Exploration Experience Research Dinner Dialogues, Women and Minority Engineering Program, NC State University, June 2024.

## EDUCATION

**B.S. Computer Science, Minor in Business Administration** **Expected Dec. 2026**

*North Carolina State University | GPA: 3.84*

**Honors**: STARS AI Scholar | Grand Challenges Scholars Program, Advance Personalized Learning
**Coursework**: Intro to AI, Privacy in the Digital Age, Software Project Management, Ethics in Computing

**Study Abroad** | Computer Science Coursework | *The Hong Kong Polytechnic University* **Spring 2026**
**Study Abroad** | Sustainable Infrastructure | Université Côte d'Azur, Nice France **Spring 2025**

### B.S. Mathematics, Magna Cum Laude — May 2022

*University at Albany, SUNY*

**Coursework**: Intro to Logic, Intro to Proofs, Probability, Linear Algebra, Statistical Inference

## HCI AND LEARNING SCIENCES RESEARCH

### Project Lead — Sep. 2026 - Present

*"Can You Learn Privately? Privacy, Access, and Personalized Learning for Tor Users", NC State University*

- Proposed an original research question and methodology from an open-ended course brief, extending a Tor .onion privacy audit into an HCI evaluation of educational access for privacy-reliant learners.
- Lead a 3-person team designing a two-rater heuristic evaluation (Nielsen and Molich) with inter-rater reliability (Cohen's kappa) to measure whether learning platforms make access blocks and privacy compromises perceivable to students who rely on Tor.

### Undergraduate Researcher (GCSP REU) — Jan. 2025 - Apr. 2025

*Game2Learn Lab (PI: Dr. Tiffany Barnes), NC State University*

- Conceived and independently developed Proof-42, a cross-platform mobile logic tutor (React Native, Tailwind CSS, Supabase), after lab survey data showed that half of students found the existing tutor's interface, not the logic, was their primary obstacle to completing proofs.
- Analyzed existing logic tutors to design a seven-level scaffolded progression grounded in cognitive-load principles, with screen-reader support and color-blind-safe palettes.
- Ran iterative peer usability testing to refine navigation and interactive proof elements.

### Undergraduate Researcher (GCSP REU) — May 2024 - Jul. 2024

*Game2Learn Lab (PI: Dr. Tiffany Barnes), NC State University | Socially Relevant Computing REU*

- Introduced bidirectional prompting (FS B-L-DCoT), which merges forward chaining from the premises with backward chaining from the goal, and evaluated it against four established chain-of-thought techniques on GPT-4 across seven Deep Thought proof problems.
- Analyzed each generated step against its predecessor and classified errors by logic rule, finding that the simplest prompting technique was most accurate, and that accuracy fell on rules involving negation and nested statements.
- Co-authored the resulting journal article [1]; credited with conceptualization, methodology, validation, and writing (original draft, review and editing).

## TEACHING, OUTREACH, AND INCLUSIVE DESIGN

### STARS AI Scholar, K-12 AI-Literacy and Accessibility Outreach — Dec. 2023 - Sep. 2025

*STARS Computing Corps, NC State University*

- Led a participatory co-design study with a blind and low-vision collaborator evaluating assistive smart glasses as a learning aid.
- Co-designed and field-tested an AI, robotics, and computational-thinking curriculum for 50+ K-12 students with disabilities and their families, evaluating scalable methods for broadening participation in computing.
- Deployed an accessible AI and robotics curriculum across a fifth-grade pilot cohort; co-authored the program's evaluation study [2].

### Teaching Assistant, Introduction to Computing (OOP and Java) **Aug. 2024 - Dec. 2024**

*NC State University*

- Mentored 100+ undergraduates in object-oriented programming, algorithm analysis, and debugging; held weekly office hours and gave structured, individualized feedback on assignments.

## SYSTEMS AND SOFTWARE ENGINEERING

### Software Engineer, Senior Design Capstone **Aug. 2026 - Present**

*Rapid Post-Earthquake Bridge Assessment Tool | NC State Earthquake Research Team and AKDOT*

- Co-engineer an earthquake assessment system (React, TypeScript, Tailwind, FastAPI, MySQL, Docker) with a 5-person team, supporting both simulated repair planning and post-event response, with offline capability for field engineers working without reliable connectivity across remote Alaska.
- Translate legacy MATLAB seismic algorithms into Python, structuring the model framework so NC State civil engineering researchers can independently tune ground-motion parameters after the capstone team leaves.

### Software Engineer Intern **Jun. 2025 - Aug. 2025**

*NASA Jet Propulsion Laboratory, Pasadena, CA*

- Built a JSON-driven configuration interface for Deep Space Network subsystem engineers, replacing siloed manual workflows with a single environment operators could use without specialist knowledge of every subsystem.
- Automated backend file generation with Python and Perl CGI scripts, removing manual terminal tasks for cross-functional operators.

### Research Scientist **May 2024 - Aug. 2024**

*NASA L'SPACE Mission Concept Academy | Interdisciplinary Lunar Mission Design*

- Co-designed a 199-page lunar water-ice detection mission concept with a nationwide, cross-disciplinary student team, defining system requirements, a Science Traceability Matrix, and a Failure Modes and Effects Analysis so specialists in different subsystems could evaluate one another's decisions.

### Classroom Technology Technician **Jul. 2023 - Jul. 2024**

*NC State University Office of Information Technology*

- Performed root-cause analysis on classroom instructional technology across campus so teaching did not stop when equipment failed.

## LEADERSHIP AND APPLIED RESEARCH

### Ventures Development and Research Lead **May 2026 - Jul. 2026**

*Associated Credit Management, Hong Kong SAR*

- Ran cross-cultural user-retention analysis of regional edtech platforms for a four-venture digital learning ecosystem, studying how local regulatory and infrastructure conditions change what a learning platform does for its users.

### Treasurer and Fundraising Chair **Aug. 2024 - Dec. 2025**

*Grand Challenges Student Organization, NC State University*

- Exceeded fundraising targets and led the organization's transition to official nonprofit status, making it eligible for grants and tax-deductible donations to continue supporting engineering students.

## HONORS AND AWARDS

**Grand Challenges Scholars Program**, Advance Personalized Learning | NC State University (2023-2026)
**GCSP Research Experience for Undergraduates**, funding Game2Learn Lab research | NC State (2024, 2025)
**NASA L'SPACE Mission Concept Academy** | national, cross-institutional cohort (2024)
**STARS Award** | STARS Computing Corps (2025)
**Robert E. Gross / Lockheed Aircraft Corporation Scholarship** | NC State College of Engineering (2025)
**Connections Award** | ASA DataFest (2025)
**Dean's List** | NC State University (2023, 2024)
**Magna Cum Laude** | University at Albany, SUNY (2022)

## RESEARCH ETHICS AND TRAINING

**CITI Program Certifications**, 2024 - 2027
Human Subjects Research, Responsible Conduct of Research, Social, Behavioral and Educational Research Good Clinical Practice, Conflicts of Interest, Technology Ethics and Regulations
**Developing Cultural Competence**, NC State University, 2024

## TECHNICAL SKILLS

**Research methods**: heuristic evaluation, inter-rater reliability, usability testing, LLM evaluation, prompt engineering
**Languages**: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS, C, Perl
**Tools & databases**: Figma, Git, MySQL, Supabase, Docker, Nginx, Postman, Apache, Expo Go
**Frameworks & libraries**: React, React Native, Next.js, Tailwind CSS, FastAPI, SQLAlchemy, Pandas, Spring, Hibernate, jQuery, Bootstrap